

Spraying Through Deformable Nozzles: Rheology Inspired by Insects

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Abstract

We perform an exploratory investigation into the effect of a nozzle’s material on the characteristics of its spray, when sprayed through with water between 0.5 and 4.0 bar. Sprays were characterised by their Weber number and spray angle, determined via OpenCV frame analysis recorded with a high-speed camera. It was observed that a nozzle manufactured from vinyl-polysiloxane (PVS) resin produced a non-zero spray angle with similar internal geometries and in comparable conditions where a rigid glass capillary needle nozzle did not, with the PVS nozzle also having a noticeably lower Weber number at most pressures, implying a significant effect of the properties of the material on the instability of the sprayed liquid jet. Additionally, we discuss the engineering and manufacturing challenges faced when working with non-rigid-materials and the challenges faced with the use of rigid material nozzles in industry.

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1 Introduction

Liquid jets and sprays are used everywhere in science and industry, such as in combustion [1], agriculture, and pharmaceuticals, as well as being essential tools for survival in nature, such as in self-defence with the bombardier beetle [2]. It is key for us to understand the physics behind the breakup of a jet into droplets, so that we may improve upon the effectiveness of their use, for example, to improve the efficiency of drug delivery in a nasal spray [3]. For this reason, the physics behind liquid jets have been studied extensively for hundreds of years, as early as in the notebooks of Leonardo da Vinci [4]. Though an intensely complex problem, involving many aspects of physics (such as non-linear dynamics and resonance), the breakup of a liquid jet into a string of droplets after a length known as the ‘breakup length’ can be attributed to linear disturbances upon the column of liquid from a perturbing force (for example, air resistance), and the surface energy minimisation of a cylindrical column of liquid driving the radius of the jet to zero [5]. This, however, is not a comprehensive list, and break up is governed by a variety of factors including nozzle geometry, or liquid viscosity and pressure. The breakup of the jet can mainly be characterised via its Weber number [6],

$$We = \rho h_0 v_0^2 / \gamma, \quad (1)$$

where ρ , h_0 , v_0 are the density, unperturbed radius and velocity of the liquid jet, and γ is the surface tension. The Weber number describes the ratio between inertial forces and surface tension for a drop ejected from a liquid jet and helps us determine which forces were mainly responsible for the breakup. The Weber number has been used to characterise a number of mechanisms for the breakup of a jet and the resulting formation of droplets, called regimes, which reflect the present forces (surface tension, inertia, etc.) and resulting structures. These regimes consist of the dripping regime, the Rayleigh and first wind induced regime [7], the second wind induced regime and the atomization regime [8], see Figure 1.

At low outlet velocity, liquid from the jet accumulates at the tip until gravity overcomes the surface tension forces of the fluid, sheering the mass of water from the nozzle as a liquid drop, this is the dripping regime. As outlet velocity increases, a cylindrically symmetric jet is formed

that later breaks down into droplets after the breakup length, characterised by quantities like the pressure, velocity or viscosity of the jet, as liquid at the end of the jet accelerates beyond the main jet column velocity, and sheers into individual droplets, known as the Rayleigh regime. As the velocity of the jet increases, air resistance acts to extenuate instabilities in the jet, causing the surface tension to break and for asymmetric separation of droplets from the jet at a reduced breakup length (first wind induced regime). Further velocity increases cause resistive effects to become even more pronounced, leading to a chaotic rupture of the jet and expulsion of droplets (second wind induced regime). The random nature of the breakup makes it hard to tell if the breakup length increases or decreases at this point (note the dotted line on Figure 1). At the highest velocities, the jet separates into droplets with a breakup length of almost zero, known as atomization. Atomised sprays are incredibly important in industry, giving the greatest aeration in fuel injection in combustion, or elsewhere in spray coating/painting [9]. Due to this high relevance as well as the complexity of the physics involved, lots of research has been done into *numerically* simulating the atomization of a cylindrical liquid jet into a stagnant gas or other relevant situations [10], in the name of making these sprays more effective.

Traditionally, engineering solutions have utilised rigid, hard materials for high strength, reliable and long-lasting components. However, one of the most common problems faced in the usage of nozzles in industrial processes is *clogging*, which can lead to reductions in efficiency and quality in the applied spray [11]. It is only recently that non-rigid materials such as polymers and liquid crystals, have been brought under the umbrella of ‘soft matter’, and utilised in the scientific frontiers, such as biomedicine or electronics [12], and it is here we seek the answer to the problem of nozzle clogging. In the natural world, many creatures, notably insects, utilise fluid sprays and jets and means of survival. In many situations, the clogging of an animal’s nozzle, and therefore it’s inability of spray effectively, could be the difference between life and death [13]. As such, in this experiment we aim to draw inspiration from evolution’s solution to this problem, in the use of non-rigid materials in the production of nozzles as a passive means of adjusting the stability of a liquid jet, and to explore how the velocity

and angular spread of a jet changes for different material nozzles at varying pressures.

2 Experimental Methods

A basic schematic of the experimental set-up used for this study can be observed in Figure 2. The water dispensing system consists of an air compressor, water tank and a nozzle connect in series with tubing, with the nozzle and water tank separated by a switch. We opted to use an air compressor in this experiment over alternative means of delivering pressure (eg a syringe pump) due to a need for a reliable constant pressure (for example, we found syringe pumps would stall quite easily). The end of the tubing connected to the nozzle has a metal mount with a screw hole in the end. The video capture system consists of a high-speed camera aimed at the outlet of the nozzle, back lit by a light source for exposure (a ring light for us, but a spotlight and diffuser plate would also work) and adjacent to a translucent ruler. The high-speed camera is connected to a computer running the camera software ‘E-Bus Player’ [14], which we used to record and save images. An example of this setup can be observed in Figure 4. The markings on the ruler must be over the light source from the camera’s point of view so that the markings may be clearly visible for calibration in later image analysis of results. Below the nozzle is a collection vessel, for example a bucket or measuring tube, so the volume of water sprayed is measured. In this study, deionized water was used as the spray fluid, as we decided it would be best to reduce risk of clogging in the nozzles from rogue debris. Water has a density (ρ) of $0.99701 \text{ g cm}^{-3}$ and surface tension (γ) of $71.97 \text{ dynes cm}^{-1}$ [15] in room temperature and pressure (what we will now refer to as ambient conditions). Our basic nozzle design consists of a hard plastic ‘base’, which can be screwed into the metal mount to connect it to the water tank, upon which a nozzle design can be attached, as discussed in Sections 2.1 and 2.2. We wrapped teflon tape around the screw of the base for its hydrophobic properties, to decrease the volume of water leakage. These bases were 3D printed, the smaller radius end threaded as a screw and then drilled through the centre with a 1 mm drill bit, to the specifications in the schematic in Figure 3, seen also is an example of a base. Initially, we performed a pressure/flow rate calibration of

the equipment with no nozzle or base inserted into the setup. In this case the collection vessel was a measuring cylinder, and a stopwatch was included in the setup. With the system compressed to a pressure between 0.5 and 3 bar (the maximum safe pressure for our system), the switch was closed, and the water discharged into the measuring cylinder for 1 to 2 seconds, timed on the stopwatch. This is then repeated multiple times at random increments of 0.5 bar between our experimental range, so that an average and standard deviation may be extracted from the results.

2.1 Glass Capillary Nozzles

The first nozzle design we tested utilised glass needles, as seen in Figure 5, to observe a spray from a nozzle made from a rigid material. The glass needles were manufactured from glass capillaries of diameter 1 mm using a Sutter Instrument P-1000 Micropipette puller (see [16]), which uses tension and an electronic heating element to melt and pull the capillary apart into 2 hypodermic needles. The insides of the capillary remains unobstructed due to capillary action, but the tip may close as the melted glass cools. As such, the tip must be broken off as close to the end as possible to reopen the needle opening, while keeping the opening diameter as small as possible. The result, when inserted into the drilled hole of the base and glued using an epoxy adhesive (we used araldite), is a long glass capillary nozzle with an opening diameter D in the order of hundreds of microns. A schematic for this design can be seen in Figure 6.

2.2 Deformable Nozzles

The second nozzle design we tested were nozzles formed by pouring a deformable resin into a metal mould. A schematic and example of the metal mould can be seen in Figure 7. We cleaned the mould with acetone to remove any residual particulates such as left over adhesive from the mould, and then lubricated the mould with a small spray of WD-40 (if needed we then used a can of compressed air to remove excess and pop any bubbles formed in the lubricant). We then threaded a length of 200 micron diameter wire through a base, and through the end of the mould, bending the wire at 90 degrees at the end of the mould so that it may sit flush atop our work surface, with the base centred above

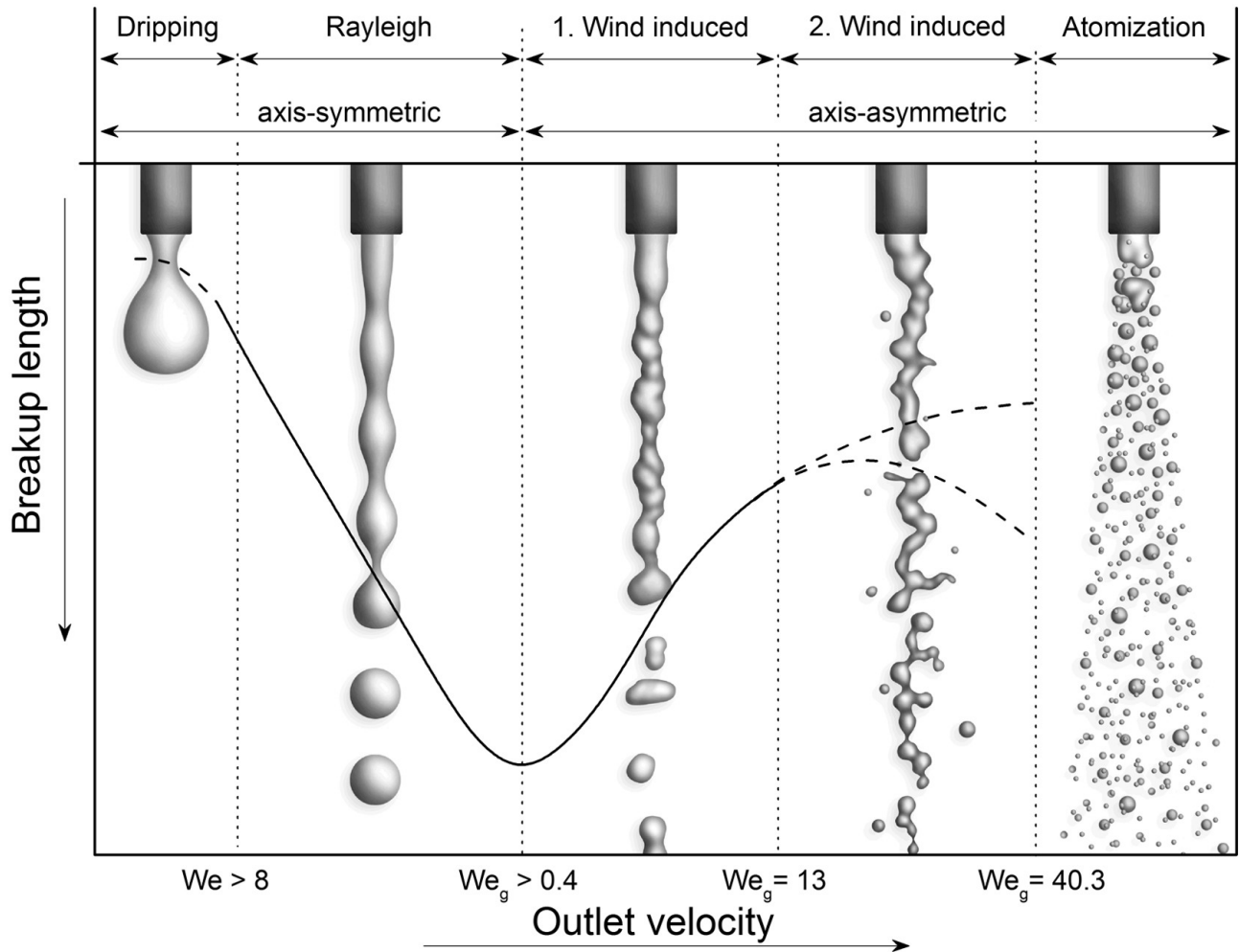


Figure 1: (Provided by Elsevier Inc.) A figure showing the breakup regimes, namely how the breakup length and characteristics change with outlet velocity. Note the change in Weber number from liquid to gas along the x-axis as ambient gas effects become significant at higher velocities.

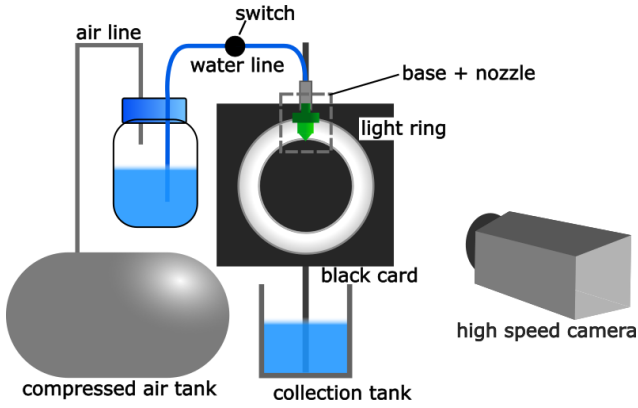


Figure 2: A schematical representation of the experimental setup. Shown is a water dispensing system, video capture system and a basic schematic of a nozzle expelling a jet of water. L denotes the length of the nozzle, L_b is the breakup length of the jet, D is the diameter of the nozzle opening and U is the output velocity of the jet.

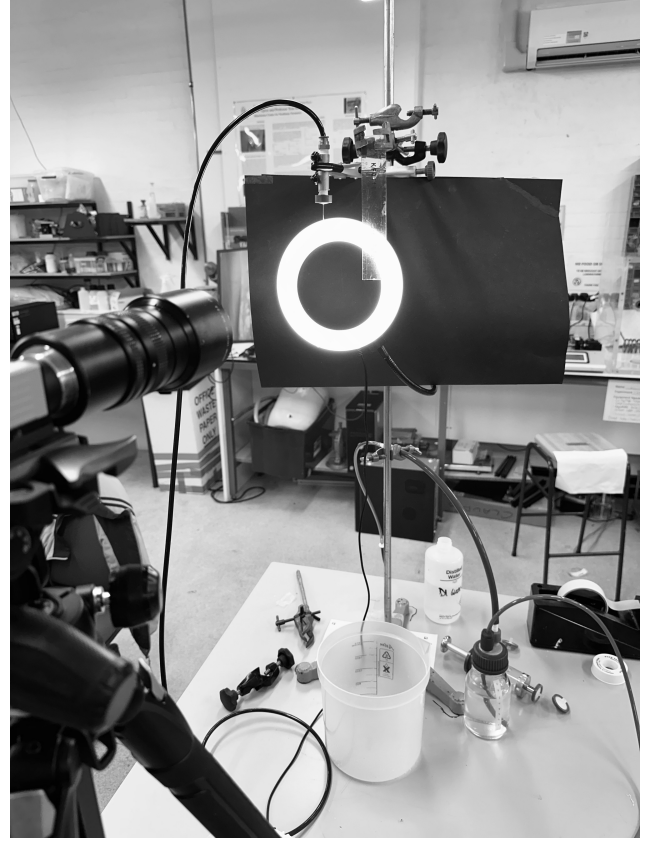


Figure 4: An example setup of our schematic, shown is our camera incident on a glass capillary nozzle (see Section 2.1), in front of a light ring and backing of black card.

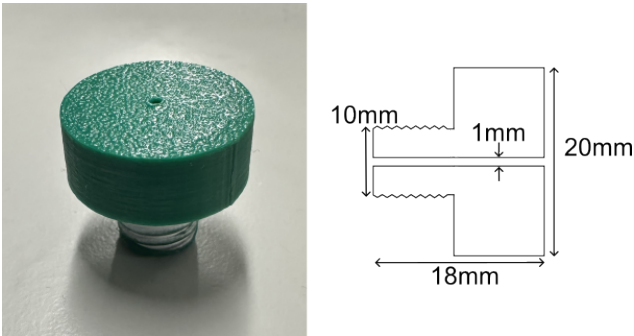


Figure 3: An example of our 3D printed plastic bases (left) along with a schematic (right). Note the teflon tape around the screw of the base in the picture.

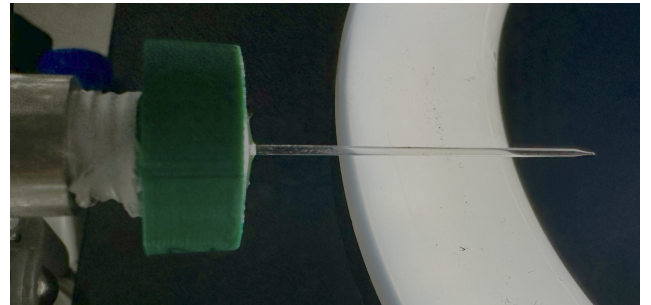


Figure 5: An image of one of our glass capillary nozzles. Shown is the glass capillary needle, glued onto the 3D printed green plastic base, which is that screwed into the metal mount of the tubing after being wrapped in teflon tape.

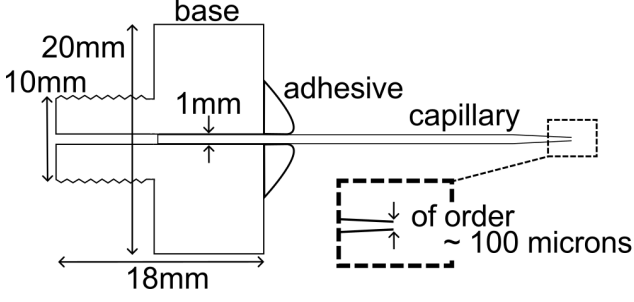


Figure 6: A schematic of our glass capillary needle nozzle design. Shown is the plastic base discussed in Section 2, along with a glass capillary needle inserted through the drilled hole of the base and glued into place with the epoxy adhesive. The opening of the capillary is shown to be in the order of 100 microns.

the mould, this then creates the internal geometry once the resin is poured in and set. For our study we worked with a vinyl-polysiloxane resin, henceforth referred to as PVS, with a measured Young's modulus (E) in the range $270 \leq E \leq 955 \pm 35$ KPa [17], depending on the mixing ratio of its base and catalyst, which we mixed in a 1:1 ratio for our experiment. We found for our mould design, 4 grams total of PVS was enough to fill the mould with a small excess. The resin mixture will start to set as soon as the base and catalyst are combined, so it must then be mixed and passed through a vacuum chamber (to remove any bubbles trapped in the mixture) within a couple of minutes to ensure the mixture does not set so much that it cannot be poured easily. Once poured, we then left the resin to set for at least 30 minutes to be allowed to fully set. We then glued the resulting nozzle to the base using the same epoxy adhesive used in Section 2.1. The result is a deformable nozzle with an opening diameter of approximately 200 microns. A schematic of this design and an example can be seen in Figure 8.

2.3 The Procedure

Once we had constructed our nozzles, we proceeded to screw them into the metal mould in our set-up. First, we adjusted the focus and exposure on the lens of the high-speed camera to make the readings on the ruler clear against the backlight, and a single image was captured for pixel-to-metre calibration in our results. Then, we turned the exposure to its maximum to make the spray as clear as possible to the camera. An example of a calibration image can be

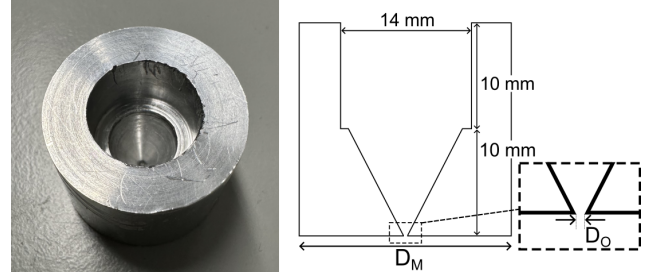


Figure 7: Shown is a schematic showing the resin mould used to manufacture deformable nozzles (right) and an example of a mould (left). The external diameter of the mould, denoted by D_M , is not significant and changes between our different moulds. The tip diameter of the mould, denoted by D_O is approximately 1 mm.

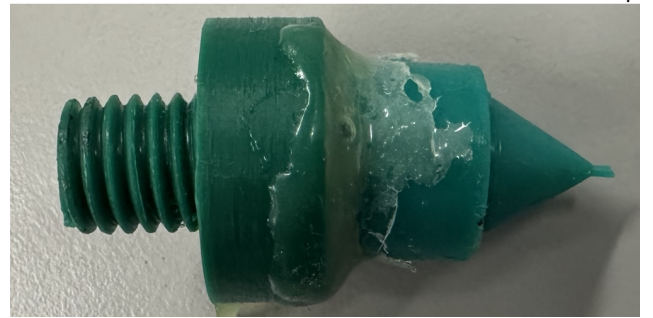
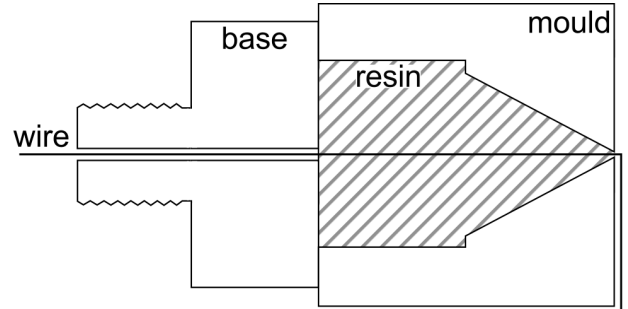


Figure 8: Shown is a schematic of our deformable nozzle manufacturing process during setting (top), along with an example of a completed nozzle, removed from the mould and glued to a base. The schematic shows a wire, of 200 micron diameter, threaded through our base centred above our mould (As discussed in Sections 2 and 2.2 respectively).



Figure 9: An example of a calibration image from our set-up, taking while testing with a deformable nozzle. Note the clear markings of the ruler against the white of the light ring.

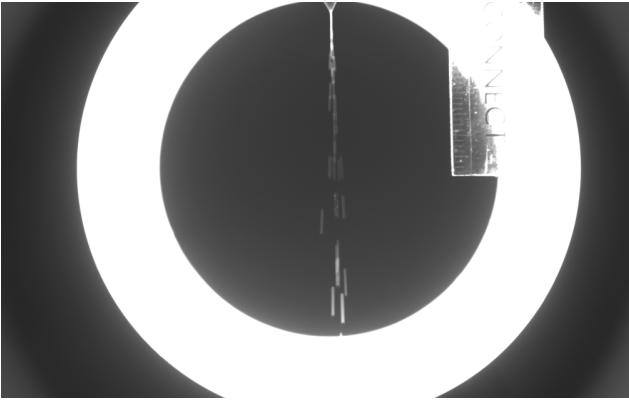


Figure 10: An example frame of a spray, taken while testing a deformable nozzle. Note the exposure being higher than in Figure 9, this is done to make the spray appear clearer against the background.

seen in Figure 9. We sprayed water through each nozzle at pressures between 0.5 and 3.0-4.0 bar depending on what the nozzle can withstand, at iterations of 0.5 bar, taking time between spans of multiple measurements to allow the compressor to equalise. The recording is started just before the switch on the system is flipped to allow the water through, in order to capture at least a couple frames of background before the spray begins. The recording is then stopped after approximately a second, to ensure the images captured are all at constant pressure (it is assumed that the pressure delivered by the air compressor may vary slightly over larger time scales), and the switch flipped to block the flow of water again. An example frame from one of our sprays can be seen in Figure 10.

3 Analysis and Discussion

3.1 Image Analysis

The data analysis on our results was done via image analysis through the OpenCV library [18] in Python, due to its high optimization, making it quick and easily scalable to cover the vast numbers of images recorded over this experiment (order 1000). First, we must calibrate the program with a scale factor to convert between pixels and metres in our image. This is done using the image of the ruler placed in the same plane as the nozzle as observed in Figure 9 (It is assumed the nozzle and the ruler are far enough from the lens that they are approximately the same distance away).

The image of the ruler is cropped to only the centimetre markings and processed using functions from the OpenCV library so that the markings on the ruler stand out as white lines on a black background. The array of values that makes up the image is then flattened using the NumPy library [20], and its peaks detected using the SciPi library [21]. The distance in pixels between the millimetre markings on the ruler was then found from the distance between the peaks on the resulting signal graph, giving the scale factor of pixels to metres. A section of this code can be seen in Figure 20, along with an example result of the code in Figure 11. The error on the scale factor comes primarily from the alignment of the ruler not being exactly in line with the axis of the camera, giving a non-zero peak width and hence a slight standard error in the peak-to-peak width. To get the velocity of a spray from its images, we took the last frame before water is observed to be exiting the nozzle as our background image, and the last frame before the tip of the spray went below the lower radius of the light ring to be our last image. We then subtracted the background image from the last frame after processing, resulting in the length the jet travelled between the two frames. We then use OpenCV contour detection to obtain the length of this shape in pixels. We can then determine the initial velocity of the jet using the frames-per-second of our camera as well as the scale factor we worked out previously, using the following formula,

$$u = \frac{L}{i\tau\alpha}, \quad (2)$$

where u is the initial jet velocity, L is the length the spray travels between the background and

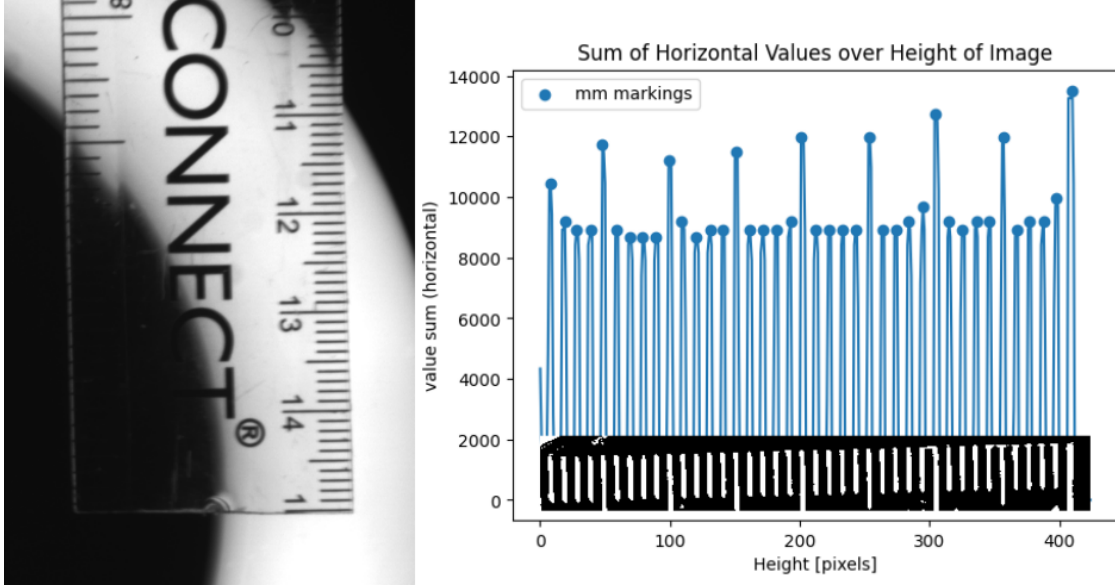


Figure 11: An example of our calibration result. Shown is a cropped picture of the reference ruler (left), and the resulting summed horizontal values over the height of the image, graphed out using the matplotlib.pyplot library [19] with peaks relating to the millimetre markings on the ruler indicated (right). Note the processed, cropped millimetre markings shown in black and white on the right figure line up with the peaks on the graph. The result of this calibration gave a scale factor of 10308 ± 285 pixels per metre.

last frames, τ is the frame time of the camera, or $1/\text{frames-per-second}$ of camera (fps), i is the number of frames between the background and last, and α is the scale factor from pixels to metres. Our code for this process can be seen in Figure 21. A figure showing the result of this algorithm can be seen in Figure 12. This process has an error that is affected heavily by the frame speed of the camera, as the actual start of the spray could be anywhere between the background frame and the first frame of the sequence. There is also a dependence on the uncertainty on the position of the forefront most drop of the spray in the last frame, due to the smearing effect caused by the fps of the cameras being too low, but this is considered negligible in comparison. Following the analysis of the capillary nozzles, we slightly altered this method to measure the path of the spray between the first frame spray is observed, and the last frame as before, removing the time dependence on the error, making the main dependence of the error on the uncertainty of the droplets position.

To get the angle of the spray, we started by taking that last frame before any water is observed exiting the nozzle to be our background and subtracting this background from every subsequent image. The differences are grayscaled and then processed using local contrast enhancement,

a bilateral filter and a binary threshold. We then superimpose every image in the sequence together to give a ‘shadow’ of the sprays total path. We then use OpenCV to detect the inner radius of the ring light and use this as a mask to avoid after artifacting from the ring light. We then used a probabilistic Hough line transform to detect the shapes of the droplets from the resulting image, which then allowed us to extract the angular bounds of the spray after filtering out especially short and faint lines. An example figure showing this result can be seen in Figure 13. The error on the spray angle comes primarily from the uncertainty of the droplets position due to the “smearing” effect due to the camera fps as discussed above, and also from the faintness of individual droplets compared to the overall shape, making determining the edges of the spray cone by the Hough lines algorithm difficult.

3.2 Pressure-Flow Rate Calibration

The results from our pressure-flow rate calibration with no nozzle inserted into the system can be seen in Figure 14. We take the error on the pressure to be half the resolution of the dial on our air compressor, which is a constant 0.1 bar. We observe that these results show an approxi-

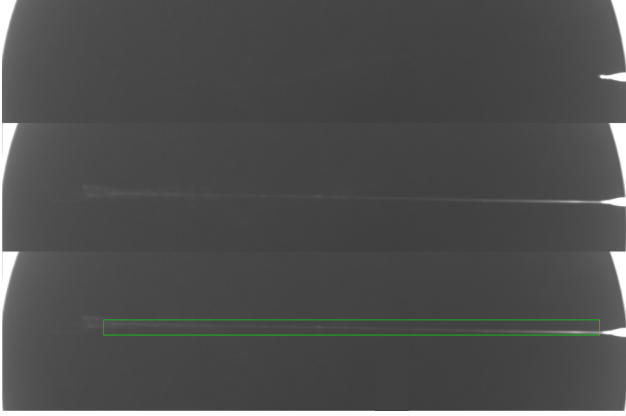


Figure 12: An example velocity calculation for a glass capillary nozzle at 3.0 bar. Shown is the background frame of the sequence (top), the last frame of the sequence (middle), and the last frame of the sequence with the bounding box of the spray's travel length in green. The velocity from this result is $6.64 \pm 0.97 \text{ms}^{-1}$, over 2 frames at 7ms per frame.

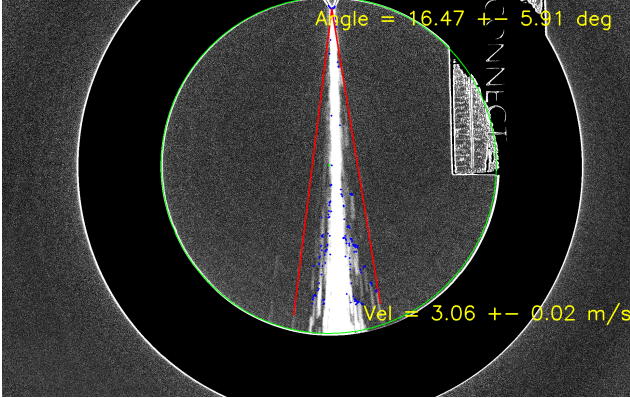


Figure 13: An example result showing the spray angle and velocity of a PVS deformable nozzle sprayed through at 3.5 bar. Shown is the spray "shadow" in the centre in white, indicating the sum paths of all droplets detected. The blue dots indicated detected individual droplets (concentrated near the edge as they cannot be detected in the centre), and the red lines indicate the bounds of the spray. The velocity is then calculated as discussed previously. The resulting angle and velocity from this figure are 16.47 ± 5.91 degrees and $3.06 \pm 0.02 \text{ms}^{-1}$ respectively.

mately linear relationship between flow rate and pressure for our system, with a moderately high but acceptable reduced chi squared. We postulate that this loose fit may be a result of variability in the pressure supplied by our compressor over continued use, highlighting the need for time to be taken between the taking of multiple results to allow the compressor pressure to equalise in proceeding measurements. Ideally for a full exploration of this trend, many more measurements would be taken to cancel this effect out on average, but for getting an idea of the approximate trend these results are suitable for our needs.

3.3 Spray Angles and Jet Velocity

Following image analysis of a PVS deformable nozzle and a glass capillary nozzle, we plot our data points in Figures 15 and 16. The first significant observation is the difference in spray angle between the two types of nozzles. For the capillary nozzles, there is *no observable spray* (see Figure 12) at any pressure, the flow is entirely laminar, whereas for the deformable nozzles, there is a clearly observable non-zero spray angle at all pressures despite comparable conditions and internal geometry to the capillary nozzles. It is unclear however if there is an overall trend due to the lack of repeated readings at the same pressures, due to issues discussed later in Section 3.4. The second observation relates to the calculated Weber number of the sprays, calculated from the initial velocities obtained from our image analysis. It is clear from Figure 15 that the Weber number of the capillary nozzle varies significantly more than the deformable nozzle, and at higher values generally. While that would imply that the capillary nozzle spray should have a more significant spray angle than the deformable nozzle (due to higher inertial force effect leading to greater instabilities), the opposite is true when looking at the observed sprays. It is worth noting the anomalously high Weber number for the two lowest pressures for the capillary nozzle (discussed in Section 3.4), with the trend otherwise increasing Weber number with pressure and hence velocity. This implies that the presence of the deformable material in the nozzle produces greater instabilities in the liquid jet, leading to greater spray angle and breakup at the orifice of the nozzle.

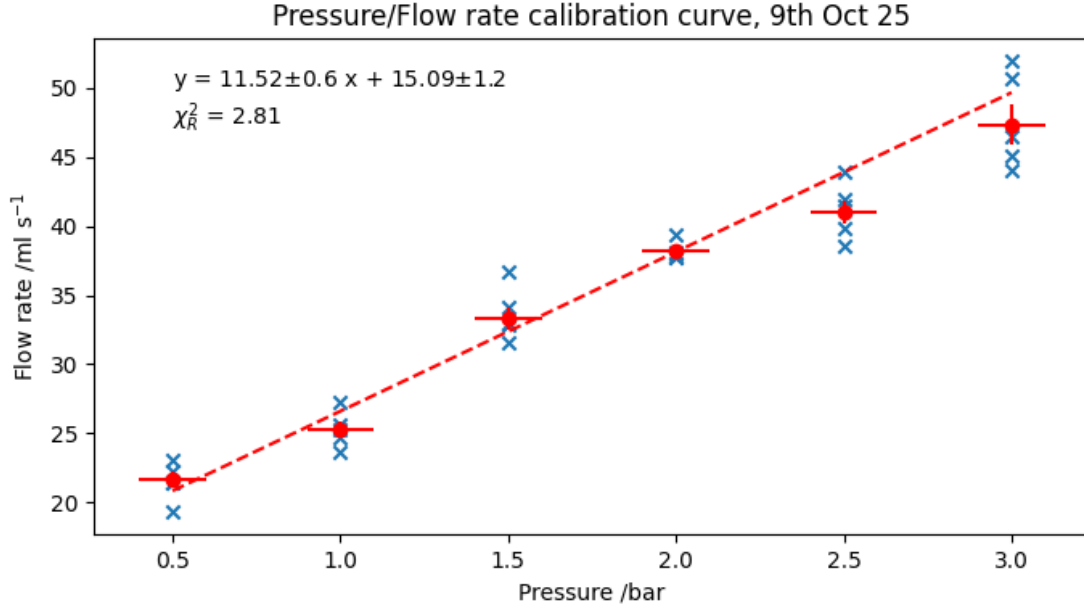


Figure 14: A graph showing the water flow rate against pressure results with no nozzle inserted into our system. On the y-axis is the flow rate of water in millilitre per second, on the x axis is the pressure in bars. The individual results at each pressure are marked by the blue crosses, while the mean flow rates with their associated error are marked in red. The linear least squares fit indicated by the red dotted line shows a gradient of $11.52 \pm 0.6 \text{ ml s}^{-1} \text{ bar}^{-1}$ with an intercept of $15.09 \pm 1.2 \text{ ml s}^{-1}$. The reduced chi squared of this fit is 2.81.

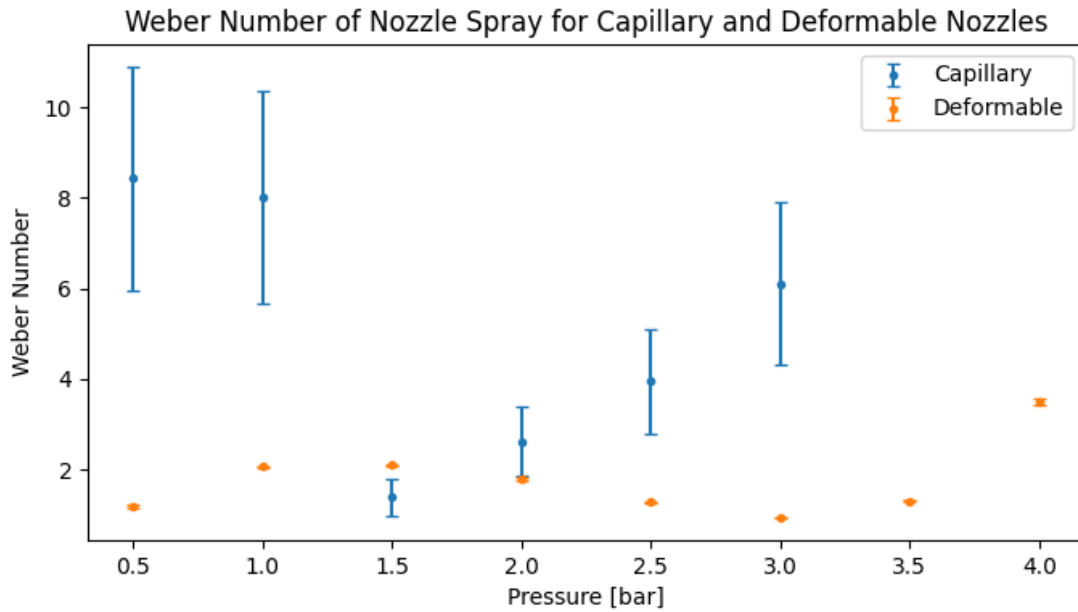


Figure 15: A figure showing the results from our image analysis for the Weber number of a deformable nozzle (orange) and rigid glass capillary nozzle (blue) spray. On the X axis is the pressure of the compressor in bar, and on the y axis is the Weber number calculated with the initial velocity obtained from image analysis.

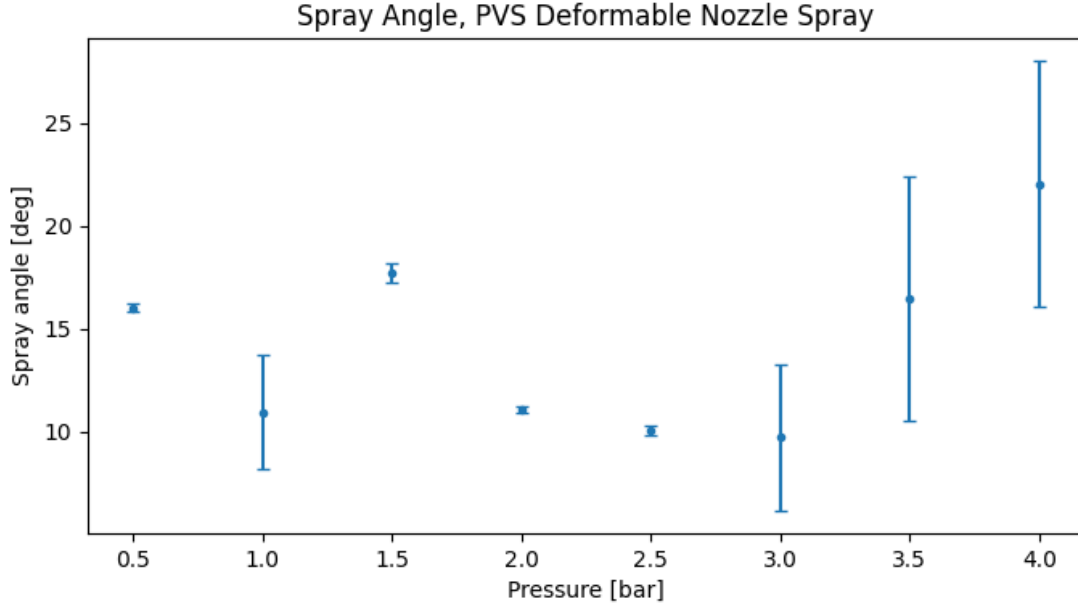


Figure 16: A figure showing the results from our image analysis for the spray angle of a deformable PVS nozzle spray, with the angles at their respective pressures being the blue points with error. On the X axis is the pressure of the compressor in bar, and on the y axis is the spray angle in degrees.

3.4 Nozzle Performance

By far the greatest challenge in this exploratory study was the performance of the nozzles. For example, we found with the capillary nozzles in particular that they would extremely readily become blocked or clogged, rendering them unable to produce a measurable spray, an example of a blocked capillary nozzle can be seen in Figure 17. We also discovered that under our highest operating pressures (3-4 bar), the capillary would not be able to withstand the pressure and would break either at the tip, widening the outlet diameter and inducing laminar flow, or breaking the tip of the needle off completely, hence the upper limit of 3 bar for the capillary nozzle. If the outlet of the capillary needle is broken off too small, this can also happen at lower pressures (0.5-1.5 bar) leading to sudden change in the outlet velocity of the nozzle. We believe this is the cause of the sudden jump in Weber number observed in Figure 15 between 1.0 and 1.5 bar for the capillary nozzle. For the deformable nozzle, we found there were significant challenges in the manufacturing of the nozzle. Due to the nature of using a liquid resin that needs time to cure, we found it was difficult to introduce any non-simplistic internal geometry into the nozzle, due to any internal moulds introduced also needing to be removed after curing, which

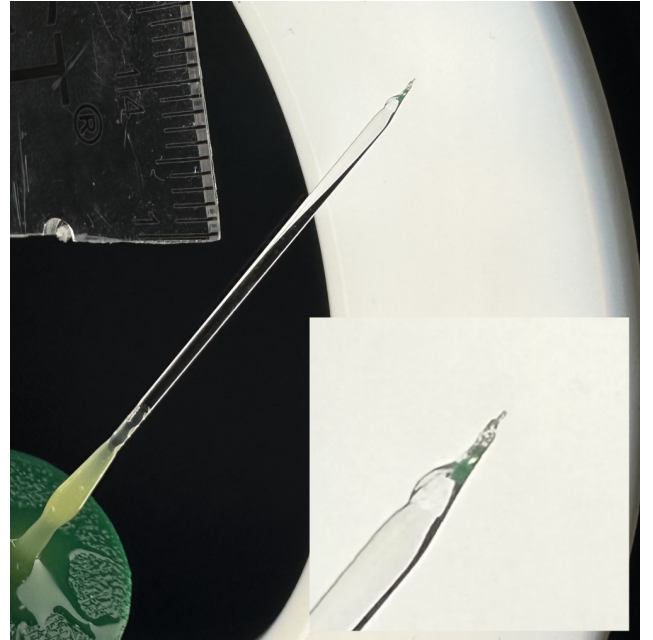


Figure 17: An example of a nozzle blocked with plastic fibres likely fragmented from the nozzle base due to the high water pressure. In the lower right is a zoomed in section of the very tip of the nozzle so the blockage can be clearly made out.



Figure 18: An image taken while testing with our deformable nozzle design, where water burst out of the epoxy adhesive at the PVS-plastic boundary between our nozzle and our base. The image is cropped to highlight the spray of water coming out of the side of the nozzle, just above the top of the image.

would make the manufacture of deformable nozzles with internal cavities such as swirling chambers or other complex internal geometries used in industry [22] difficult. It is for this reason we settled on the simple centred wire design specified in Section 2.2, and the matching internal geometry of the rigid glass capillary nozzle. Another challenge we faced when developing the deformable nozzles was at the PVS-plastic base boundary. At this point, the radius of the central channel suddenly changes from the 1 mm diameter of the base to the 200 micron diameter of the wire, resulting in a large pressure differential. We found that the PVS resin would not adhere well enough to the epoxy resin, resulting in the water bursting out from the nozzle at this point, as seen in Figure 18, with the resulting broken nozzle seen in Figure 19.

4 Conclusion and Future Work

We found that a nozzle manufactured from a vinyl-polysiloxane resin in a 1:1 base-catalyst ratio produced a clear observable spray where a rigid glass capillary nozzle produced only a thin laminar jet under comparable conditions and internal geometries. We also found the Weber number for the deformable nozzle was generally lower than for the capillary, implying a contribution from the soft material on the instability of the liquid jet. We also discussed the challenges faced when working with rigid nozzles, and with the manufacturing and spraying of deformable nozzles. Unfortunately, due to the time con-



Figure 19: The broken nozzle resulting from the bursting that occurred in Figure 18. The nozzle is clearly separated from the base, with clear gaps in the adhesive around the boundary where water could spray out.

straints of the experiment, these challenges were not able to be fully explored and faced, leading to a lesser number of repeat results than we would like. In future work, we would like to see a more robust and reliable nozzle design created, particularly to tackle the issue of the PVS to rigid plastic weak point in the design, so that a full quantitative exploration may be performed. We would also see multiple different resins of different hardness/Young’s modulus tested in the same nozzle design for a full overview on the effect of the nozzle’s material on its spray characteristics, along with a much higher fps camera so that individual droplets may be resolved and a distribution of droplet sizes may also be obtained.

```

1 ruler = cv2.imread(imPath + '/' + images[0]) # Grabs the image of the ruler
2 mean_img = cv2.pyrMeanShiftFiltering(ruler,20,30) # Smooths the image
3 gray_img = cv2.cvtColor(mean_img, cv2.COLOR_BGR2GRAY) # Applies a gray scale
4 th = cv2.adaptiveThreshold(gray_img,255,cv2.ADAPTIVE_THRESH_MEAN_C,cv2.
5   THRESH_BINARY_INV,11,2) # Apply a harsh threshold
6 th2 = th[round(ruler.shape[0]*0.1):round(ruler.shape[0]*0.45),round(ruler.shape
   [1]*0.84):round(ruler.shape[1]*0.87)] # crop the image to the markings
7
8 projection = np.sum(th2, axis=1) # sum horizontal values
9 peaks, properties = sp.signal.find_peaks(projection, height=4000) # get peaks of
   resulting signal
10 scale_factor = (peaks[-1]-peaks[0])/(len(peaks)-1) #[pixels mm^-1], *1000 for
   pixels m^-1
11

```

Figure 20: A segment of our calibration code. The program first reads the image of the ruler, and passes it through a mean filter to smooth any noise. The image is then gray-scaled and passed through a binary inverted threshold, resulting in the black ruler markings being left as white pixels and everything else as a black background. We then crop the image in line 6. Then, we sum the axis along the ruler markings, and find the distance between the resulting signal's lowest and highest peak (shown graphed on the right in Figure 11) divided by the number of peaks for the scale factor in pixels per mm, in lines 9 and 10. Here, 'imPath' refers to the directory where the images are contained, and 'np' refers to the NumPy library.

```

1
2 # Process images
3 first = cv2.imread(filePath + "/" + images[0])
4 last = cv2.imread(filePath + "/" + images[-1])
5 firstGray = Gray(first)
6 lastGray = Gray(last)
7 difference = cv2.subtract(lastGray, firstGray)
8 differenceProcessed = ProcessImage(difference)
9
10 # mask to remove any effects from the lightring etc to only the rough bounds of the
    spray
11 mask = np.zeros_like(differenceProcessed)
12 mask = cv2.rectangle(mask, (round(mask.shape[1]*0.45), 30), (round(mask.shape
    [1]*0.65), round(mask.shape[0]*0.95)), (255,255,255), -1)
13 differenceProcessed = cv2.bitwise_and(differenceProcessed, mask)
14
15 # Find contours
16 contours, hierarchy = cv2.findContours(differenceProcessed, cv2.RETR_TREE, cv2.
    CHAIN_APPROX_SIMPLE)
17 contoursSorted = sorted(contours, key=lambda x: cv2.contourArea(x))
18 # Filter Contours to a minimum area (removes anomolous effects)
19 minArea = 350
20 contoursSorted = [contour for contour in contours if cv2.contourArea(contour) >
    minArea]
21
22 # find bounding boxes
23 boundingBoxes = []
24 for contour in contoursSorted:
25     x, y, w, h = cv2.boundingRect(contour)
26     boundingBoxes.append([x,y,x+w,y+h]) # store 4 corners of bounding box
27 boundingBoxes = np.array(boundingBoxes) # convert to np array
28
29 x_min = np.min(boundingBoxes[:,0]) # get the bounds
30 x_max = np.max(boundingBoxes[:,2]) # of all detected
31 y_min = np.min(boundingBoxes[:,1]) # bounding boxes
32 y_max = np.max(boundingBoxes[:,3]) #
33
34 distPixels = y_max - y_min
35 time = TIMESTEP * (len(images)-1)
36 dist = distPixels/scale
37 velocityM = dist/time
38

```

Figure 21: An algorithm to determine the initial velocity of a spray, using image analysis. The program first grabs and process the background and last image of the sequence, determined either manually or through further analysis, and subtracts the background from the last (lines 2 through 8). Then we mask the result to the approximate bounds of the spray to remove any residual contours from the light ring (lines 11-13). We then use OpenCV contour detected to detect the spray path, filtering out any small effects from noise (lines 16-20). We then find the bounding box of all remaining contours, and process this length into the initial velocity of the spray in metres per second (lines 23-37).

References

- [1] G. M. FAETH, L.-P. HSIANG, and P.-K. WU, “Structure and breakup properties of sprays,” *International Journal of Multiphase Flow*, vol. 91, 1995.
- [2] T. Eisner and D. J. Aneshansley, “Spray aiming in the bombardier beetle: Photographic evidence,” *Proceedings of the National Academy of Sciences*, vol. 96, no. 17, pp. 9705–9709, 1999.
- [3] K. Inthavong, Z. Tian, J. Tu, W. Yang, and C. Xue, “Optimising nasal spray parameters for efficient drug delivery using computational fluid dynamics,” *Computers in Biology and Medicine*, vol. 38, no. 6, pp. 713–726, 2008.
- [4] L. da Vinci, *Codex Leicester*. Sydney, N.S.W.: Powerhouse Museum, 2000. Facsimile of Leonardo’s manuscript treatise on water dynamics.
- [5] J. Eggers and E. Villermaux, “Physics of liquid jets,” *Reports on Progress in Physics*, vol. 71, 2008.
- [6] Weber and Constantin, “Zum Zerfall eines Flüssigkeitsstrahles,” *Zeitschrift Angewandte Mathematik und Mechanik*, vol. 11, pp. 136–154, Jan. 1931.
- [7] P. K. Kirar, N. Kumar, and K. C. Sahu, “Dynamics of jet breakup and the resultant drop size distribution: effect of nozzle size and impingement velocity,” *Indian Institute of Technology Hyderabad*, 2024.
- [8] B. Bonhoeffer, A. Kwade, and M. Juhnke, “Impact of formulation properties and process parameters on the dispensing and deposition of drug nanosuspensions using micro-valve technology,” *Journal of Pharmaceutical Sciences*, vol. 106, 2017.
- [9] E. Trautner, J. Hasslberger, S. Ketterl, and M. Klein, “Primary atomization of liquid jets: Identification and investigation of droplets at the instant of their formation using direct numerical simulation,” *International Journal of Multiphase Flow*, vol. 160, p. 104360, 2023.
- [10] M. Herrmann, “On simulating primary atomization using the refined level set grid method,” *Atomization and Sprays*, vol. 21, no. 4, 2011.
- [11] D. Meng, S. B. Nandipati, A. K. Silaen, Y. Wang, S. Abraham, D. Brown, and C. Zhou, “Impacts of cooling reduction due to spray nozzle clogging on shell formation in continuous casting of steel,” *Metals*, vol. 15, no. 10, 2025.
- [12] P. Lang and Y. Liu, *Soft Matter at Aqueous Interfaces*. Springer Cham, 2015.
- [13] P. A. Rühs, J. Bergfreund, P. Bertsch, S. J. Gstöhl, and P. Fischer, “Complex fluids in animal survival strategies,” *Soft Matter*, vol. 17, pp. 3022–3036, 2021.
- [14] P. Technology, “ebus player, ebus sdk version 5.0 user guide,” 2017.
- [15] Zumdahl and S. S, “Water,” *Encyclopedia Britannica*, 2025. Accessed 11 November 2025.
- [16] A. Oesterle and Sutter Instruments, “Pipette cookbook 2018, p-97 p-1000 micropipette pullers,” 2018.
- [17] S. Monnery, S. Jain, C. Johnson, D. Pihler-Puzović, and F. Box, “Impacting metamaterials: The threshold for wave propagation in holey columns,” *Physical Review Materials*, vol. 8, 07 2024.
- [18] G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, 2000.
- [19] J. D. Hunter, “Matplotlib: A 2d graphics environment,” *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

- [20] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant, “Array programming with NumPy,” *Nature*, vol. 585, pp. 357–362, Sept. 2020.
- [21] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.
- [22] R. Alidoost Dafsari, F. Vashahi, and J. Lee, “Effect of swirl chamber length on the atomization characteristics of a pressure-swirl nozzle,” *Atomization and Sprays*, vol. 9, pp. 859–874, 01 2017.